CERTAIN-COAST — Text for the NATO SPS application form

Multi-Year Project · call 2027-1 · every field in the order of the official form · apply on https://natosps.grantplatform.com

How to use this document

• **This is the only document you need for the online form.** Go through it from top to bottom and copy each answer into the field with the same name on the platform. The other CERTAIN-COAST documents are background and are not uploaded.

• **Word limits are respected:** the green line under each long answer gives its length.

• **Yellow text** is information only you can give (names, dates, signatures, choices). Replace it before submitting.

• **Uploads and signatures** are listed in the checklist below; all uploads are PDF files.

• **Deadline:** 15 November 2026, 23:59 CET (call 2027-1, checked on nato.int on 6 November 2026). The call asks for full proposals in one round: fill every field now. Limits per field are in words; there is no page limit except one page per institution overview.

Checklist: uploads and signatures

Upload	Format	Who signs or prepares it	Status
SPS MYP Acceptance of Funds (NATO template)	PDF	Filled in: CERTAIN-COAST_SPS_MYP_Acceptance_of_Funds.docx; the NPD and a representative of the NPD's institution (Head of Institution or Financial Authority) sign; upload in tab 4	[to do]
SPS MYP Agreement for Joint Research (NATO template)	PDF	Filled in: CERTAIN-COAST_SPS_MYP_Agreement_for_Joint_Research.docx; all Co-Directors and a representative of each institution sign	[to do]
SPS MYP Detailed Budget workbook (NATO Excel template, tabs "Info" and "Detail")	Excel	Fill by hand (the form warns that using AI on this file may corrupt it and lead to automatic rejection), typing the values of CERTAIN-COAST_Budget_Workbook_Guide.docx cell by cell	[to do]
Project schedule with a Gantt chart	PDF	Ready: CERTAIN-COAST_Project_Schedule.pdf	ready
End-user commitments, all letters merged into one file	PDF	Each end-user on its own letterhead (drafts in the Letters of Commitment document)	[to do]
Brief overview of participating institutions (at most one page each)	PDF	Draft: CERTAIN-COAST_Institutions_Overview.docx; fill the yellow fields, save as PDF	[to do]
Financial rules of the NPD's institution	Link	Ready: the UPT Statute in English (Chapter IV, Articles 24–25), https://upt.edu.al/wp-content/uploads/2023/02/Statuti-PDF-English.pdf. If the UPT finance office provides its internal regulation on administration, finance and accounting, upload that PDF instead	ready
Intellectual Property Rights agreement or waiver	PDF	Draft: CERTAIN-COAST_IPR_Agreement_Draft.docx; all Co-Directors, their institutions and the end-users sign; recommended now (listed in the 2027-1 call), allowed after approval	[to do]
Illustrative figures (optional, one PDF with a table of contents)	PDF	Ready: CERTAIN-COAST_Illustrative_Figures.pdf (6 pages, 12 figures)	ready
Additional files (optional): technical annex, later with the CVs	PDF	Ready: CERTAIN-COAST_Technical_Annex.pdf (7 pages: methods, preparatory tests, gates and fallbacks, the six new contributions). When the CVs are filled in, they are merged with it into one PDF with a table of contents	ready

Still missing before you can submit: [a partner-country Project Director, employed and resident in an eligible partner country] (without one the application is not eligible); [confirmation of the Albanian institution of the NPD]; [signed letters from the five end-users].

1. Start here

Project title

About 100 characters; understandable to non-experts; link to security and NATO.

Certified early warning to protect water, port and energy infrastructure on a sinking Adriatic coast

100 characters

Project short title

80 characters; must match cell E2 ("Short Title") in the "Info" tab of the SPS MYP Detailed Budget workbook.

CERTAIN-COAST

Confirmation that you have read the Multi-Year Project Handbook

[Tick after reading the latest MYP Handbook.]

How did you hear about this opportunity?

[Your choice: word of mouth, journal, conference, online, SPS Information Day or other.]

Is the proposal a resubmission of a previous unsuccessful application?

[No] (unless you applied with this project before).

Did you use any generative AI tools in the preparation of your research proposal?

Yes. Parts of the text were drafted with an AI assistant and checked by the applicants. Answer honestly.

2. Eligibility

Are the Co-Directors (including NPD and PPD) affiliated with a governmental, academic, or other non-profit institution?

Yes. [Both directors must be employed by a university, public institute or other non-profit body.]

Do any of the Co-Directors (including NPD and PPD) hold other ongoing SPS grants?

[No] (an individual may hold only one SPS grant at a time).

Is the application developed jointly by one or more Co-Directors from NATO countries and one or more from eligible NATO partner countries?

Yes. [Requires the partner-country Project Director.] Then click "Check Eligibility".

3. Activity Info

Primary Key Priority

Select one in the dropdown.

Environment, Climate Change and Security

Additional key priorities (optional, at most three)

Resilience; Critical Underwater Infrastructure; Innovation and Emerging Disruptive Technologies (EDTs)

What security challenge(s) would your project address?

500 words.

Three kinds of critical infrastructure of a NATO Ally stand on the same low Adriatic coastal plain of Albania, at two sites. At Durrës lie the wellfields that supply drinking water and the Port of Durrës, from which 93.6% of Albania's seaborne freight left in the second quarter of 2025 (INSTAT). Further south, near Fier, the Trans Adriatic Pipeline (TAP) leaves the shore for its 105 km crossing to Italy, carrying gas from Azerbaijan to Europe. The plain is built of soft sediments that settle slowly (Métois et al.,

2020); at Durrës the ground is sinking close to active thrust faults, and the Mw 6.4 earthquake of 26 November 2019 caused liquefaction along the coast.

The assets face three linked hazards. Seawater intrudes into the aquifer and can make wells unusable for years. In an earthquake, saturated sand can liquefy under quays, foundations and the landfall. Storm surges and extreme sea levels flood low ground. Land subsidence and sea-level rise worsen all three at once, because they raise the sea and the water table together. Pumping moves the hazards in opposite directions: it draws seawater towards the wells but lowers the water table and so strengthens the ground against liquefaction. Managing one hazard alone can increase another.

Today these hazards are monitored by different institutions with different models, and no warning states how often it misses. For security decisions this is the critical gap: an operator who must cut pumping, restrict a quay or inspect the landfall needs to know how far a warning can be trusted, and civil protection needs one consistent picture of water, port and energy after an earthquake or a storm.

The security challenge is therefore to keep three essential services of an Ally working, and to know in advance when they will stop being safe. The project serves three of NATO's seven baseline requirements for national resilience (resilient food and water resources, resilient civil transportation systems and resilient energy supplies) and the topic of particular interest of the 2027-1 call, "protection of critical infrastructure, supplies and personnel; maritime and undersea infrastructure resilience". NATO's 2022 Strategic Concept calls climate change "a crisis and threat multiplier"; on this coast the multiplication is physical and measurable.

A second challenge is trust in machine-learned tools used in security decisions. CERTAIN-COAST uses artificial intelligence only with a certified error bound and a miss rate that is checked in operation, in line with NATO's Principles of Responsible Use of AI.

409 words (limit 500)

What technical challenge(s) would your project address, if any?

500 words.

Four technical problems stand between current science and a warning that operators can act on.

1. The hazards are coupled but modelled apart. Seawater intrusion, aquifer compaction, liquefaction and coastal water levels depend on the same two variables, relative sea level and the water table, yet are computed by separate models. A pumping decision cannot be checked against all of them at once.

2. Relative sea level must be known at each asset, not only at the tide gauge. The land at Durrës moves vertically by different amounts from place to place, including the 2019 coseismic step; this motion must be measured with satellite radar (InSAR) and GNSS and carried into every hazard model with its uncertainty.

3. Physical models are too slow for warnings, and machine-learned copies come without a guarantee. Warnings need thousands of model runs a day. Neural-operator surrogates make this possible, but a surrogate used for safety must state its error for each forecast.

4. No warning in this field states how often it misses. Recent statistical methods (online conformal prediction under distribution shift, guarantees that hold within regimes such as droughts, extreme conformal prediction for events rarer than the record, and anytime-valid verification) can provide such guarantees, but they have not been applied to coupled coastal hazards driven by slow physics, with dependent alarms and few extreme events.

Three further problems have no published solution. There is no method to compute the set of pumping and dewatering decisions that keeps all the hazards within certified limits at once, nor the sea level or year at which this set disappears; tipping points exist for seawater intrusion alone, not for the coupled system. And monitoring networks are designed to reduce today's forecast errors; there is no criterion for how many years of safe operation a sensor buys. And routine pumping never shows how the aquifer responds near the onset of intrusion, which decides how long safe operation lasts; we found no pumping-test design that reveals it safely and values the test in years.

All of this must work on a real coast with incomplete records, under security restrictions on data, and in a form that the operators can run themselves after the project.

366 words (limit 500)

What will the project deliver or develop by the end of the grant?

500 words; show how it advances the state of the art.

By Month 36 the project delivers:

1. A coastal motion atlas from Durrës to Fier and relative sea level at each asset, from Sentinel-1 InSAR since 2014, reflectors and GNSS (D1.1).

2. A monitoring network chosen for the warnings and handed over to the operators (D4.2): piezometers with conductivity loggers, a barometer and sealed hub channels that correct them, a local rain gauge, GNSS, seismic nodes and, where access allows, fibre-optic sensing. Each sensor is bought only if its guaranteed value (band removed, years of safe operation added) is positive.

3. One calibrated model that couples groundwater flow and salt, aquifer compaction, liquefaction under a moving water table and coastal water levels (D2.1), and a fast neural-operator copy with a certified error bound (D2.2); a generative copy explores long storylines. The inputs and boundaries nobody measures are reconstructed from rain and heads, with bands valid when the simulator or rain product is wrong or the climate drifts.

4. Certified alarms for seawater intrusion, liquefaction and extreme water levels whose miss rate is bounded in advance (indicatively 5–10%), with a joint guarantee for the three assets, back-tested since 2014 under a pre-registered protocol (D3.1, D3.2).

5. Decisions for the water utility: the effect of each wellfield's pumping on salinity with the hidden drought demand removed; the salinity under drier recharge; the certified safe operating window and the year it closes; safe pumping tests that sharpen it; and a pumping rule agreed with the utility.

6. Twelve months of live bulletins with the five end-users, every outcome logged and the miss rates verified (D5.1); operating procedures and a policy brief (D6.2); open software and data with a DOI (D6.1).

How this advances the state of the art. The alarms apply conformal and anytime-valid methods of 2023–2026 to coupled coastal hazards. Six new contributions were tested before the project. (i) Guarantees for quantities nobody measures: inputs and boundaries known only through rain, calibrated on simulated aquifers and measured anchor periods, with a computable floor on how narrow each band can be. (ii) The value of a sensor as the guaranteed band it removes and the years of safe operation it adds (in tests, a tide gauge rated useful by a standard ensemble widened the band by 16–21%). (iii) Pumping effects free of hidden confounding: reading the recharge from the heads removed the bias of drought-driven pumping that history-based substitutes kept. (iv) The certified safe operating window and safe pumping tests chosen for the years they add. (v) A network that corrects its own sensors: reference channels remove air-pressure and logger-drift errors, and the robust design has a certified distance from the best network (+52% worst-case information in tests). (vi) Warnings for what happens between samples: the chance that chloride crosses its limit at any time in the season, as a certified band over all models that fit the records (34 times the chance of an exceedance on a sampling date in tests). The integrated system reaches TRL 6 on site, with the operators.

500 words (limit 500)

How will the project reach the planned end state?

500 words; methodology, approach and how the teams collaborate.

Seven work packages run over 36 months in three phases, separated by evidence gates with thresholds fixed in advance.

Observe and measure (Months 1–12). WP4 designs the network before buying it: each candidate sensor (piezometer with conductivity logger, rain gauge, GNSS, seismic node, barometer and sealed hub channels that correct the loggers for air pressure and drift) is ranked by the guaranteed warning band it removes and the years of safe operation it adds per euro; the network is chosen against the worst plausible sensor error, with a certified optimality gap (D4.1, Month 6); a logger or samples paired a few days apart are added where sparse samples leave crossings between them open. WP1 obtains

vertical land motion and relative sea level at each asset from Sentinel-1 InSAR since 2014 and GNSS, and the head limit for pumping from aquifer compaction.

Model and certify (Months 4–30). WP2 builds one coupled model in stages (flow and salt, then compaction and liquefaction). The inputs and boundaries nobody measures are reconstructed from rain and heads by a generative prior with an ensemble update. Their bands are calibrated on simulated aquifers and on measured anchor periods, so they stay valid if the simulator is wrong; under drift the calibration is matched to the forecast horizon, and a computable floor shows how far each band is from the narrowest possible. Gate MS3 (Month 18) requires held-out heads, chloride, InSAR and seismic velocity inside the 90% intervals. A neural-operator copy with a certified error bound then serves the warnings. WP3 agrees thresholds and tolerable miss rates with the operators, builds the alarms (crossings between samples included, as a certified band over the models that fit the records), back-tests them since 2014 under a protocol registered before the results; gates MS4 (Month 22) and MS5 (Month 27) decide live operation and confirm the miss rates.

Operate, verify and hand over (Months 22–36). WP5 runs the system live for twelve months, logs every bulletin with its outcome, verifies the miss rates with anytime-valid tests and runs a post-event protocol. It estimates what each wellfield's pumping does to the salinity with the hidden drought demand removed: the heads, corrected for the measured drawdown, read the recharge, and a calibrated check rejects any substitute that misses a confounder. It computes the salinity under drier recharge, the certified safe operating window, its closing year and safe pumping tests that sharpen it, and agrees a pumping rule with the utility. WP6 trains early-career researchers and operators, releases open software and transfers the method to a partner coast. WP7 manages data access, security classification, risks and reporting.

Collaboration. The Albanian team leads WP1–WP5 and WP7; the partner-country team leads WP6, a transfer study on its own coast and a user-side review of the bulletins. Early-career researchers of both countries work across work packages, with two-month exchanges. End-users set thresholds, join a dry run and co-sign operational milestones; a steering committee with them meets every six months, and each gate has a named fallback.

500 words (limit 500)

May your proposal raise any ethical issue?

Yes, limited: personal data of workshop and training participants, and security-sensitive data about critical infrastructure. If a text box appears, use:

CERTAIN-COAST involves no animals, human tissue or clinical research. Its ethical issues are people and their data, security-sensitive information about critical infrastructure, machine learning in safety decisions, field work and the responsibility of issuing warnings. Participants in workshops, interviews and tests of the bulletins take part voluntarily with informed consent, after institutional ethics approval; personal data are minimised, pseudonymised and protected under Albania's Law No. 124/2024, aligned with the GDPR. Asset-level vulnerability results are shared only with operators and authorities through a release gate. The machine-learned surrogate follows NATO's six principles of responsible AI: every forecast carries an error bound, every bulletin is logged and reproducible, and operators keep the decision. Each bulletin states its certified miss rate, the miss-false-alarm trade-off is agreed with the operators in advance, public warnings remain with the authorities, and the back-testing protocol is frozen before results are seen. An independent ethics advisor and a yearly review oversee these measures.

State-of-the-art: Related Projects / Duplication

Related research outside the project team (2020–2026):

- Coastal groundwater under climate change and pumping. Rising sea level lifts the water table and extends flooding inland (Befus et al., 2020); 250,000 US coastal wells show where heads lie below sea level (Jasechko et al., 2020); by 2100 recharge sets the severity of intrusion and sea level its prevalence (Adams et al., 2024); ensemble pumping optimisation handles parameter and climate uncertainty (Coulon et al., 2024); machine-learned surrogates predict intrusion fast (Jiang, Liu and

Dwivedi, 2025). These give scenarios without guarantees and treat pumping as known and exogenous.

• Inputs and boundaries from sparse data. Ensemble and diffusion-based inversion (Zheng et al., 2025; Yao et al., 2025), satellite and machine-learning recharge (Soylu, Entekhabi and Bras, 2026) and evaluations of gridded rain products (Abbas et al., 2026) estimate what is not measured, without a calibrated guarantee and without carrying the rain-product errors into the boundaries.

• Guarantees for forecasts and surrogates. Conformal risk control (Angelopoulos et al., 2024), online calibration under shift (Gibbs and Candès, 2024; Feldman et al., 2023), robustness to distribution shift (Aolaritei et al., 2025), conformal neural operators (Ma et al., 2024), extreme conformal prediction (Pasche, Lam and Engelke, 2026), the expected size of conformal sets (Dhillon, Deligiannidis and Rainforth, 2024) and anytime-valid inference (Ramdas et al., 2023) all calibrate on observed outcomes; none covers a boundary that is never observed, asks how narrow such a guarantee can be, or how long it holds under drift.

• Monitoring design. Bayesian optimal design places sensors for PDE-governed inverse problems (Alexanderian, 2021; Wu, Chen and Ghattas, 2023); greedy selection is near-optimal for submodular criteria (Krause, Singh and Guestrin, 2008), which correlated errors break (Liu et al., 2016; Uciński, 2020; Attia and Constantinescu, 2022); robust designs maximise the worst case over an uncertain error model (Attia, Leyffer and Munson, 2023; Chowdhary, Attia and Alexanderian, 2025); recent work places several sensor types (Maio et al., 2026), designs tsunami warning networks (Venkat, Henneking and Ghattas, 2026) or ranks wells by predictive skill (Meray et al., 2022). None values a sensor by the guaranteed band it removes or the years of safe operation it adds, or certifies a robust network's distance from the best one; all take correlated errors as given, although air pressure and logger drift are routinely removed with a nearby barometer (Rasmussen and Crawford, 1997; Toll and Rasmussen, 2007), outside the design.

• Forecasting from irregular records and robust bounds. Neural jump ODEs learn the optimal online forecast from irregular, incomplete records (Herrera, Krach and Teichmann, 2021; Krach, Nübel and Teichmann, 2026); robust finance bounds a path-dependent price over all models that fit the liquid prices (Hobson, 1998; Beiglböck, Henry-Labordère and Penkner, 2013), computed with neural SDEs (Gierjatowicz et al., 2020). Neither gives a finite-record guarantee or names the observation that would narrow the bound. Yet sampling pins the marginal law of chloride, not its time scale, so a crossing of the limit between samples is bounded only from below (Slepian, 1962).

• Hidden confounding over time. The Time Series Deconfounder infers substitutes for hidden confounders from many treatments (Bica, Alaa and van der Schaar, 2020); proxy methods (Mastouri et al., 2021), synthetic interventions (Agarwal, Shah and Shen, 2026), conformal intervals for individual effects (Alaa, Ahmad and van der Laan, 2023), policy counterfactuals (McKay and Wolf, 2023), causal dose curves (Singh, Xu and Gretton, 2024) and adjoint attribution in the ocean (Fukumori, Wang and Fenty, 2021) complete the picture. Substitutes built from the treatments cannot see the fresh news of a confounder such as a heat wave that makes every farmer pump, and none uses a physical measurement of the confounding path.

• Vertical land motion and coastal exposure from InSAR and GNSS (Ohenhen et al., 2023), and slow settling of the Albanian coastal plain near Fier (Métois et al., 2020); the 2019 Durrës earthquake and its liquefaction (Mavroulis, Lekkas and Carydis, 2021).

What none of these does. No study combines intrusion, liquefaction and extreme water levels in one model with warnings whose miss rate is stated in advance; guarantees the inputs known only through rain when the simulator is wrong; values a sensor by the guaranteed band it removes and the years it adds; places the reference channels that correct the sensors; certifies crossings between samples; reads the hidden drought demand from the heads; or computes when a safe way of operating the assets ends. A final check in bibliographic databases will be made before submission.

Related activities of the project participants: [list proposed, ongoing or recently completed projects of the Co-Directors on related topics, with dates and budgets, or write "none"]. The preparatory technical work of the NPD's team in 2026 (eleven reviews with tests, a calibrated chloride model of the Fushë-Kuqe aquifer) is the starting point of the project, not a separately funded activity.

Other activities at the site known to the applicants: [list any national, EU or donor projects on subsidence, the port or the water supply of Durrës, or write "none known"].

What makes CERTAIN-COAST stand out: one model for three hazards at three critical assets, warnings and pumping decisions whose error is stated in advance and verified in operation, and a certified planning horizon for the operators.

References:

Abbas, A., Yang, Y., Pan, M., et al. (2026). Comprehensive global assessment of 24 gridded precipitation datasets across 18 428 catchments using hydrological modeling. Hydrology and Earth System Sciences 30(11), 3399–3423. https://doi.org/10.5194/hess-30-3399-2026

Adams, K. H., Reager, J. T., Buzzanga, B. A., David, C. H., Sawyer, A. H., Hamlington, B. D. (2024). Climate-induced saltwater intrusion in 2100: recharge-driven severity, sea level-driven prevalence. Geophysical Research Letters 51(22), e2024GL110359. https://doi.org/10.1029/2024GL110359

Agarwal, A., Shah, D., Shen, D. (2026). Synthetic interventions: extending synthetic controls to multiple treatments. Operations Research 74(2), 840–859. https://doi.org/10.1287/opre.2025.1590

Alaa, A. M., Ahmad, Z., van der Laan, M. (2023). Conformal meta-learners for predictive inference of individual treatment effects. Advances in Neural Information Processing Systems 36. https://proceedings.neurips.cc/paper_files/paper/2023/hash/94ab02a30b0e4a692a42ccd0b4c55399-Abstract-Conference.html

Alexanderian, A. (2021). Optimal experimental design for infinite-dimensional Bayesian inverse problems governed by PDEs: a review. Inverse Problems 37(4), 043001. https://doi.org/10.1088/1361-6420/abe10c

Angelopoulos, A. N., Bates, S., Fisch, A., Lei, L., Schuster, T. (2024). Conformal risk control. 12th International Conference on Learning Representations (ICLR 2024). https://arxiv.org/abs/2208.02814

Aolaritei, L., Wang, Z. O., Zhu, J., Jordan, M. I., Marzouk, Y. (2025). Conformal prediction under Lévy–Prokhorov distribution shifts: robustness to local and global perturbations. Advances in Neural Information Processing Systems 38. https://arxiv.org/abs/2502.14105

Attia, A., Constantinescu, E. (2022). Optimal experimental design for inverse problems in the presence of observation correlations. SIAM Journal on Scientific Computing 44(4), A2808–A2842. https://doi.org/10.1137/21M1418666

Attia, A., Leyffer, S., Munson, T. (2023). Robust A-optimal experimental design for Bayesian inverse problems. arXiv:2305.03855. https://arxiv.org/abs/2305.03855

Befus, K. M., Barnard, P. L., Hoover, D. J., Finzi Hart, J. A., Voss, C. I. (2020). Increasing threat of coastal groundwater hazards from sea-level rise in California. Nature Climate Change 10(10), 946–952. https://doi.org/10.1038/s41558-020-0874-1

Beiglböck, M., Henry-Labordère, P., Penkner, F. (2013). Model-independent bounds for option prices—a mass transport approach. Finance and Stochastics 17(3), 477–501. https://doi.org/10.1007/s00780-013-0205-8

Bica, I., Alaa, A. M., van der Schaar, M. (2020). Time Series Deconfounder: estimating treatment effects over time in the presence of hidden confounders. Proceedings of the 37th International Conference on Machine Learning, PMLR 119. https://proceedings.mlr.press/v119/bica20a.html

Chowdhary, A., Attia, A., Alexanderian, A. (2025). Robust optimal experimental design of infinite-dimensional Bayesian nonlinear inverse problems. arXiv:2409.09137. https://arxiv.org/abs/2409.09137

Coulon, C., White, J. T., Pryet, A., Gatel, L., Lemieux, J.-M. (2024). An ensemble-based approach for pumping optimization in an island aquifer considering parameter, observation and climate uncertainty. Hydrology and Earth System Sciences 28(1), 303–319. https://doi.org/10.5194/hess-28-303-2024

Dhillon, G. S., Deligiannidis, G., Rainforth, T. (2024). On the expected size of conformal prediction sets. Proceedings of AISTATS 2024, PMLR 238, 1549–1557. https://proceedings.mlr.press/v238/dhillon24a.html

Feldman, S., Ringel, L., Bates, S., Romano, Y. (2023). Achieving risk control in online learning settings. Transactions on Machine Learning Research. https://arxiv.org/abs/2205.09095

Fukumori, I., Wang, O., Fenty, I. (2021). Causal mechanisms of sea level and freshwater content change in the Beaufort Sea. Journal of Physical Oceanography 51, 3217–3234. https://doi.org/10.1175/jpo-d-21-0069.1

Gibbs, I., Candès, E. J. (2024). Conformal inference for online prediction with arbitrary distribution shifts. Journal of Machine Learning Research 25(162), 1–36.

Gierjatowicz, P., Sabate-Vidales, M., Šiška, D., Szpruch, L., Žurič, Ž. (2020). Robust pricing and hedging via neural SDEs. Working paper, SSRN 3646241. https://ssrn.com/abstract=3646241

Herrera, C., Krach, F., Teichmann, J. (2021). Neural jump ordinary differential equations: consistent continuous-time prediction and filtering. International Conference on Learning Representations (ICLR 2021). https://arxiv.org/abs/2006.04727

Hobson, D. G. (1998). Robust hedging of the lookback option. Finance and Stochastics 2(4), 329–347. https://doi.org/10.1007/s007800050044

Jasechko, S., Perrone, D., Seybold, H., Fan, Y., Kirchner, J. W. (2020). Groundwater level observations in 250,000 coastal US wells reveal scope of potential seawater intrusion. Nature Communications 11, 3229. https://doi.org/10.1038/s41467-020-17038-2

Jiang, S., Liu, C., Dwivedi, D. (2025). GeoFUSE: an efficient surrogate model for seawater intrusion prediction and uncertainty reduction. Water Resources Research 61(9), e2024WR038898. https://doi.org/10.1029/2024WR038898

Krach, F., Nübel, M., Teichmann, J. (2026). Optimal estimation of generic dynamics by path-dependent neural jump ODEs. Stochastic Processes and their Applications. https://doi.org/10.1016/j.spa.2026.105058

Krause, A., Singh, A., Guestrin, C. (2008). Near-optimal sensor placements in Gaussian processes: theory, efficient algorithms and empirical studies. Journal of Machine Learning Research 9, 235–284. https://jmlr.org/papers/v9/krause08a.html

Liu, S., Chepuri, S. P., Fardad, M., Maşazade, E., Leus, G., Varshney, P. K. (2016). Sensor selection for estimation with correlated measurement noise. IEEE Transactions on Signal Processing 64(13), 3509–3522. https://doi.org/10.1109/TSP.2016.2550005

Ma, Z., Pitt, D., Azizzadenesheli, K., Anandkumar, A. (2024). Calibrated uncertainty quantification for operator learning via conformal prediction. Transactions on Machine Learning Research. https://openreview.net/forum?id=cGpegxy12T

Maio, S., Alexanderian, A., Koval, K., Nicholson, R. (2026). Multi-type sensor placement for PDE-based Bayesian inverse problems. arXiv:2607.08074. https://arxiv.org/abs/2607.08074

Mastouri, A., Zhu, Y., Gultchin, L., Korba, A., Silva, R., Kusner, M., Gretton, A., Muandet, K. (2021). Proximal causal learning with kernels: two-stage estimation and moment restriction. Proceedings of the 38th International Conference on Machine Learning, PMLR 139, 7512–7523. https://proceedings.mlr.press/v139/mastouri21a.html

Mavroulis, S., Lekkas, E., Carydis, P. (2021). Liquefaction phenomena induced by the 26 November 2019, Mw = 6.4 Durrës (Albania) earthquake and liquefaction susceptibility assessment in the affected area. Geosciences 11(5), 215. https://doi.org/10.3390/geosciences11050215

McKay, A., Wolf, C. K. (2023). What can time-series regressions tell us about policy counterfactuals? Econometrica 91(5), 1695–1725. https://doi.org/10.3982/ECTA21045

Meray, A. O., Sturla, S., Siddiquee, M. R., Serata, R., Uhlemann, S., et al. (2022). PyLEnM: a machine learning framework for long-term groundwater contamination monitoring strategies. Environmental Science & Technology 56(9), 5973–5983. https://doi.org/10.1021/acs.est.1c07440

Métois, M., Benjelloun, M., Lasserre, C., Grandin, R., Barrier, L., Dushi, E., Koçi, R. (2020). Subsidence associated with oil extraction, measured from time series analysis of Sentinel-1 data: case study of the Patos-Marinza oil field, Albania. Solid Earth 11, 363–378. https://doi.org/10.5194/se-11-363-2020

Ohenhen, L. O., Shirzaei, M., Ojha, C., Kirwan, M. L. (2023). Hidden vulnerability of US Atlantic coast to sea-level rise due to vertical land motion. Nature Communications 14, 2038. https://doi.org/10.1038/s41467-023-37853-7

Pasche, O. C., Lam, H., Engelke, S. (2026). Extreme conformal prediction: reliable intervals for high-impact events. Extremes 29(2), 129–155. https://doi.org/10.1007/s10687-026-00536-9

Ramdas, A., Grünwald, P., Vovk, V., Shafer, G. (2023). Game-theoretic statistics and safe anytime-valid inference. Statistical Science 38(4), 576–601. https://doi.org/10.1214/23-STS894

Rasmussen, T. C., Crawford, L. A. (1997). Identifying and removing barometric pressure effects in confined and unconfined aquifers. Ground Water 35(3), 502–511. https://doi.org/10.1111/j.1745-6584.1997.tb00111.x

Singh, R., Xu, L., Gretton, A. (2024). Kernel methods for causal functions: dose, heterogeneous and incremental response curves. Biometrika 111(2), 497–516. https://doi.org/10.1093/biomet/asad042

Slepian, D. (1962). The one-sided barrier problem for Gaussian noise. Bell System Technical Journal 41(2), 463–501. https://doi.org/10.1002/j.1538-7305.1962.tb02419.x

Soylu, M. E., Entekhabi, D., Bras, R. L. (2026). Integrating satellite retrievals, numerical models, and machine learning for global groundwater recharge estimation. Water Resources Research 62(3), e2025WR040312. https://doi.org/10.1029/2025WR040312

Toll, N. J., Rasmussen, T. C. (2007). Removal of barometric pressure effects and earth tides from observed water levels. Ground Water 45(1), 101–105. https://doi.org/10.1111/j.1745-6584.2006.00254.x

Uciński, D. (2020). D-optimal sensor selection in the presence of correlated measurement noise. Measurement 164, 107873. https://doi.org/10.1016/j.measurement.2020.107873

Venkat, S., Henneking, S., Ghattas, O. (2026). Sensor placement for tsunami early warning via large-scale Bayesian optimal experimental design. arXiv:2604.08812. https://arxiv.org/abs/2604.08812

Wu, K., Chen, P., Ghattas, O. (2023). A fast and scalable computational framework for large-scale high-dimensional Bayesian optimal experimental design. SIAM/ASA Journal on Uncertainty Quantification 11(1), 235–261. https://doi.org/10.1137/21M1466499

Yao, J., Mammadov, A., Berner, J., Kerrigan, G., Ye, J. C., Azizzadenesheli, K., Anandkumar, A. (2025). Guided diffusion sampling on function spaces with applications to PDEs. Advances in Neural Information Processing Systems 38 (NeurIPS 2025). https://arxiv.org/abs/2505.17004

Zheng, H., Chu, W., Wang, A., Kovachki, N., Baptista, R., Yue, Y. (2025). Ensemble Kalman diffusion guidance: a derivative-free method for inverse problems. Transactions on Machine Learning Research. https://arxiv.org/abs/2409.20175

Funding requested from NATO SPS (in EUR)

Must equal cell G15 of the "Summary" tab of the SPS MYP Detailed Budget workbook.

382,500

4. NPD & PPD

Fill these fields directly on the platform, once for the NATO-country Project Director (NPD) and once for the partner-country Project Director (PPD):

Fields	What to enter
First name, family name, date of birth, sex, title	[personal data]
Nationality (and other nationalities)	[NPD: e.g. Albania; PPD: a NATO or eligible partner country]
Telephone, mobile, personal and work e-mail; online profile (optional)	[contact data]
Current job title; name, country and postal address of the institution	[NPD: institution in a NATO country; PPD: institution in an eligible partner country]
Employment history, education (three rows each)	[dates, positions, employers; dates, degrees, universities]
Publications and honours (optional, up to three each)	[recent and relevant to the project]

Fields	What to enter
Research focus and expertise (500 words)	[how the director's research relates to the project: e.g. seawater intrusion and groundwater modelling, coastal geodesy, geotechnics, statistics]
Other grants held (optional)	[dates, donor, grant name, short description, results]
SPS Acceptance of Funds (NPD only)	Upload the signed PDF of CERTAIN-COAST_SPS_MYP_Acceptance_of_Funds.docx (official NATO template, filled in): signature of the NPD and of a representative of the NPD's institution

5. Co-Directors (optional)

Not required. If you add a Co-Director, for example [an additional NATO-country co-director], they must be at a different institution and have their own budget in the Excel workbook. The Institute of Geosciences belongs to UPT, like the NPD's faculty, so its staff take part as participants, not as a Co-Director. Otherwise skip this tab.

6. Project Participants

Roles of Co-Directors and Participants (add rows as needed; all must be nationals of NATO or eligible partner countries):

First name	Family name	Affiliation	Position	Nationality	Task(s)	% time
[First name]	[Family name]	[NPD institution]	NATO-country Project Director	[Albanian]	Coordination; WP1 coastal motion; WP4 monitoring network; WP5 live operation; WP7 management	[30%]
[First name]	[Family name]	[PPD institution]	Partner-country Project Director	[nationality]	WP6 training, transfer and dissemination; transfer study on a partner-country coast; user-side review of the bulletins	[20%]
[First name]	[Family name]	[Institute of Geosciences, UPT]	Researcher	[Albanian]	Seismic, GNSS and hydrometeorological data; WP1, WP4	[15%]
To be recruited	(4 people)	NPD and PPD institutions	Early-career researchers (stipends)	[NATO or partner]	InSAR and GNSS; modelling; alarms; network and live operation (WP1–WP5)	[50%]
[First name]	[Family name]	[Durrës water utility]	End-user representative	[Albanian]	Thresholds (T3.1), dry run, live bulletins, pumping rule	[5%]
[First name]	[Family name]	[Water Resources Management Agency (AMBU)]	End-user representative	[Albanian]	Thresholds (T3.1), monitoring design (T4.1), pumping and drought rule (T5.3)	[5%]
[First name]	[Family name]	[Durrës Port Authority]	End-user representative	[Albanian]	Thresholds (T3.1), dry run, live bulletins	[5%]
[First name]	[Family name]	[Trans Adriatic Pipeline AG (TAP), Albania branch]	End-user representative	[nationality]	Thresholds (T3.1), dry run, live bulletins	[5%]
[First name]	[Family name]	[National Civil Protection Agency]	End-user representative	[Albanian]	Post-event protocol; live bulletins	[5%]

7. Project plan

Project abstract

200 words; understandable to non-experts.

On Albania's Adriatic coast, the drinking-water wells and port of Durrës and the landfall of a gas pipeline to Italy near Fier stand on low ground, which at Durrës is sinking, close to active faults. Three dangers threaten them: seawater entering the wells, ground that turns almost liquid in an earthquake, and the sea flooding the land in storms. The dangers are linked, because sinking land and rising seas worsen all three and pumping that protects against one can increase another, but today they are watched separately and no warning says how often it can be wrong.

CERTAIN-COAST will watch them together. Satellites, GNSS and new ground sensors will feed one computer model of the coast, copied by a fast artificial-intelligence model whose error is stated for every forecast. Each warning will state in advance how often it may miss a dangerous event, and this will be checked in twelve months of live operation with the water utility, the water agency, the port, the pipeline operator and civil protection. The project will show how many years a safe way of operating the three assets remains, and sharpen that estimate with safe pumping tests. Local operators will run the system afterwards.

200 words (limit 200)

Security relevance and real-world applications

Link to security. The project protects three essential services of a NATO Ally (drinking water, a port that handles most of the country's seaborne freight, and the landfall of the Trans Adriatic Pipeline near Fier) and serves three of NATO's seven baseline requirements for national resilience: resilient food and water resources, resilient civil transportation systems and resilient energy supplies, which also underpin civil support to the military. It falls under the 2027-1 topic "protection of critical infrastructure, supplies and personnel; maritime and undersea infrastructure resilience": the TAP landfall is where an undersea gas pipeline between two Allies, Albania and Italy, leaves the shore.

Use cases:

- Water utility: cut or shift pumping before chloride reaches drinking-water wells (weeks to months ahead).
- Water Resources Management Agency (AMBU): use the monitoring design, the chloride early warning for each supply well and the pumping and drought rule in its monitoring and in support of water-use permits for the aquifer.
- Port authority: restrict loads or operations and set inspection priorities after an earthquake; prepare for storm surges (hours to days ahead).
- Pipeline operator (TAP): inspect, protect or isolate the landfall near Fier when the ground or the water level threatens it.
- Civil protection: one consistent bulletin for water, port and energy after an earthquake or a storm.
- National authorities and resilience planners: the certified year from which no operating plan keeps the three assets safe together, and the sensors and measures that delay it most.

Each bulletin states its certified miss rate, so decision-makers know how far it can be trusted. The method uses free satellite data and transfers to other subsiding coasts of Allies and partners with wellfields, ports and energy landfalls.

Current status

The approach has been tested in preparatory work by the NPD's team (2026), on synthetic coastal aquifers and on the calibrated chloride model of the Fushë-Kuqe aquifer, which supplies Durrës:

- Boundaries known only through rain: on 500 simulated aquifers with four loggers, the raw ensemble missed 17–20% of the weeks; the calibrated bands met the 10% target (0.095–0.100).
- A wrong simulator: bands calibrated on simulations broke under a 35-day vadose delay, shifted parameters or a rain bias (risk 0.11–0.83); calibrated on 10–40 measured anchors they stayed at or below 10% under every law tested. At least 9 anchors are needed for an average guarantee and 22 for a 90% certificate.
- Drift: horizon-matched bands kept the risk at 0.003–0.082 up to 20 years ahead under every drift tested; with 30 anchor years a band exists only up to 17 years ahead.
- The rain product and the floor: a local gauge with one logger identified the recharge coefficient better (error 0.045) than the product with eight loggers (0.108); with 20 anchors the guaranteed band was 1.34–1.54 times the narrowest possible.
- Value of a sensor: a tide gauge that a standard ensemble said would narrow the recharge band by 15% widened the guaranteed band by 16–21%; the sign of each guaranteed value was predicted for 36 of 36 instrument–goal pairs.

- Counterfactual salinity at Fushë-Kuqe (calibrated model): with 40% less recharge the shallow area above 250 mg/L chloride would grow from 31 to 65 km² (effect +33 km², conformal band 15–49 km²), with coverage 0.887–0.900 on 300 simulated test sets.
- Pumping and hidden drought demand: in a test bed with eight wells, the summed effect of pumping on salinity (true value 0.94) was overstated by 0.59 without adjustment, by 0.56 with a substitute built from the pumping history (as in the Time Series Deconfounder) and by 0.00 with the recharge read from the heads; the physics check had size 6% and power 99–100%.
- Alarms and operating window: an alarm with an extreme-aware margin kept the share of seasons above the chloride limit at 4–9% across storylines, where a static alarm allowed 13–21%; the certified safe operating window closed after 38 years, while plans that watch one hazard believed they had 43–97 years; safe pumping tests chosen for the years they add raised the certified closing year from 10.8 to 31.9 years (true value 45) with chloride below 123 mg/L in 95% of runs; the anytime-valid test wrongly flagged a correctly calibrated alarm in 2.2% of runs, against 68% for re-checking a 95% bound after every event.
- Sensors that interfere: a 12-sensor network with a barometer and hub reference channels kept 52% more worst-case information than the classical design, certified within 18% of the best; ignoring the interference let 90% intervals cover only 49–62% of cases, the correction 88–93%.
- Crossings between samples: chloride with an 8-day time scale crossed 250 mg/L within a season with probability 0.78, 34 times the exceedance on a sampling date; quarterly samples left this anywhere in 0.18–1, a daily logger in 0.76–0.80; a conformal margin restored 90.3% coverage for skewed chloride (74% without).

These are tests on synthetic coasts and on the existing chloride model with illustrative values, not results at Durrës; the project calibrates and verifies them on site.

Starting data: Sentinel-1 radar images since 2014, tide-gauge records, the wells and chemical records of the water utility, and the mapped liquefaction of the 2019 earthquake (Mavroulis et al., 2021).

Team: [summarise the Co-Directors' prior research on seawater intrusion, coastal geodesy, geotechnics and statistics, with 3–5 key publications].

Impact

500 words; quantitative where possible.

If the project succeeds, the following will change at Durrës, at the TAP landfall near Fier and beyond:

- Three hazards covered by one warning with a stated miss rate, where today there is none: seawater intrusion at the wellfields, liquefaction at the port and the landfall, and extreme water levels at all three.
- Each alarm misses dangerous events no more often than its certified rate (indicatively 5–10%), shown by back-testing over the record since 2014 and checked in twelve months of live operation; the joint miss rate for the three assets is bounded by the sum of the rates.
- Each operator knows in advance the minimum value of the warning for its own decisions, computed from the certified miss rate and the cost-loss ratio agreed with it.
- The water utility operates its wells within a certified safe window, sharpened by safe pumping tests on its own wells, which lowers the risk of salinising wells that take years to recover and avoids new permanent subsidence; planners know the year from which the window closes and which measures extend it.
- The Water Resources Management Agency (AMBU) gains a monitoring design for its decisions and a chloride early warning for each supply well, in support of its water-use permits for the aquifer that supplies Durrës, with no procurement, ownership or maintenance obligations.
- At least 20 operator and civil-protection staff (at least two per operator) trained and running the system; at least six early-career researchers trained in the NATO and partner countries.
- A monitoring network of piezometers, GNSS, seismic stations and corner reflectors handed over to the operators, with a running cost of about €10,900 per year carried by them under a maintenance agreement.

• Open software and data with a DOI, installed and reproduced by a group outside the consortium; at least six open-access papers; operating procedures adopted for each asset; a policy brief for the national authorities.

• The method tested on a second coast in the partner country and ready for other subsiding coasts of the Adriatic and the Mediterranean.

For NATO, the result is a tested capability for three baseline requirements of resilience at an Allied site, and a worked example of artificial intelligence used in a security decision chain with testing, evaluation, verification and validation built in.

385 words (limit 500)

Technology Readiness Level (TRL)

TRL 6: Technology demonstrated in relevant environment

Technology Readiness Level (TRL) description

Start: TRL 3. Each component has been shown in preparatory synthetic tests (alarm margins, safe operating window, verification, guaranteed reconstruction from rain, value of a sensor, pumping effects with the hidden confounding removed, counterfactual salinity on the calibrated chloride model of Fushë-Kuqe, a network that corrects its own sensors, warnings for crossings between samples), and published methods exist for each element.

End: TRL 6, as the 2027-1 call asks (results at TRL 3–6). The integrated system (network, coupled model, certified surrogate, alarms, pumping rule and safe operating window) is demonstrated in its relevant environment, the operators' own sites, through twelve months of live operation with the water utility, the water resources agency (AMBU), the port authority, the pipeline operator (TAP) and civil protection, before the operators take it over. Seismic velocity as a coastal water-table sensor and the pumping rule with its safe pumping tests are demonstrated at the site but not yet run for a full year.

Data exploitation plan

Data required and existing sources: Copernicus Sentinel-1 radar images since 2014 (free and open); GNSS records; the Durrës tide-gauge record of the State Authority for Geospatial Information (ASIG); heads, pumping and chloride records of the water utility; groundwater level, chemistry and abstraction records of AMBU; geotechnical records of the port and of TAP at its landfall; records of existing national seismic stations; meteorological and wave data.

New data: continuous heads, conductivity and temperature in new and existing wells; GNSS and corner-reflector time series at the assets; continuous seismic noise from stations and temporary nodes, and fibre-optic sensing where access is agreed; laboratory chloride and major ions; the bulletin log with outcomes.

Collection and management: a data management plan (D7.1, Month 3) following the FAIR principles; standard formats (miniSEED and StationXML for seismic data, NetCDF with CF conventions for time series) and DataCite metadata; storage on the partners' servers with backup; open release of derived products and software with DOIs on Zenodo.

Risks and mitigation: (1) access to operator data: letters of commitment now, signed access agreements by Month 3, and satellite and public data as a fallback; (2) security of critical-infrastructure information: every output is classified before release, asset-level vulnerability, thresholds and sensor locations inside sites stay with the operators and authorities, and open products are aggregated or anonymised; (3) privacy: personal data are limited to workshop and training participants, collected with informed consent, minimised and pseudonymised under Albania's Law No. 124/2024 on personal data protection, aligned with the GDPR; (4) data quality: quality flags, loggers calibrated against laboratory chloride, and record errors carried as uncertainty in the model.

Work-plan

Sequencing: Phase 1, observe and measure (Months 1–12); Phase 2, model and certify (Months 4–30); Phase 3, operate, verify and hand over (Months 22–36). Gates MS3 (Month 18), MS4 (Month 22) and

MS5 (Month 27) decide whether each next step starts. Every guarantee is checked on simulated test sets and on held-out field data before it is used.

WP1, Coastal motion and relative sea level (Months 1–36; NATO-country team). Steps: Sentinel-1 InSAR time series since 2014 (T1.1); corner reflectors and GNSS at the assets (T1.2); trajectory model including the 2019 earthquake and the relative sea level at each asset (T1.3); elastic and inelastic compaction and the head limit for pumping (T1.4). Risk: loss of radar coherence on soft ground; mitigation: corner reflectors and scatterers on structures. Milestone: MS2 (Month 12).

WP2, One coupled model and its certified fast copy (Months 4–24; NATO-country team). Steps: model set-up for the two sites, the Durrës wellfields and port and the TAP landfall near Fier, on one coastal motion field (T2.1); coupling of flow, salt, compaction and liquefaction (T2.2); calibration (T2.3): the rain reaches the aquifer through a stochastic soil-water and vadose-delay model learned from irregular head records; the inputs and boundaries nobody measures (recharge coefficient, vadose delay, storage, sea level, inland inflow, initial head, dispersion width) are reconstructed by a function-space diffusion prior with an ensemble update through the exact solver; the errors of the rain product are simulated in the prior; bands are calibrated by conformal risk control on simulated aquifers and checked on measured anchor periods; neural-operator surrogate with a certified error bound (T2.4). Risk: the model fails validation; mitigation: build in stages and fall back to two-dimensional sections with a wider error bound. Risk: the anchors are too few (at least 9 are needed for an average guarantee, 22 for a 90% certificate); mitigation: anchors from the record since 2014 and from the new loggers, and a wider simulation-calibrated band reported as such. Milestones: MS3 (Month 18), MS4 (Month 22).

WP3, Certified multi-hazard alarm (Months 13–34; NATO-country team). Steps: thresholds and tolerable miss rates with the operators (T3.1); alarm margins and proofs (T3.2), with anchor calibration so that the guarantee does not depend on the simulator, horizon matching under drift, and the floor that shows how far each band is from the narrowest possible, and warnings for crossings between samples: the chance that chloride crosses its limit at any time in the season, as a band over all models that fit the records, widened by a conformal margin calibrated on simulated aquifers and anchor periods, and issued online as the conditional probability given the irregular samples and the daily heads; joint guarantee and allocation of tolerances (T3.3); back-testing (T3.4); verification in operation (T3.5). Risk: too many false alarms; mitigation: acceptable rates agreed in advance and tolerances allocated by consequence. Risk: drift faster than the record allows; mitigation: report the longest horizon for which a band exists. Risk: chloride more skewed than the model family; mitigation: the conformal margin, re-estimated each season. Milestone: MS5 (Month 27).

WP4, Monitoring network designed for the warnings (Months 1–36; NATO-country team). Steps: design (T4.1), in which each candidate sensor is valued by the guaranteed band it removes, its floor and the years of safe operation it adds per euro, with a local rain gauge among the candidates and one logger next to the wellfield most exposed to local demand; the monitoring wells of AMBU (existing wells only) are candidate sites and the design also serves AMBU's monitoring decisions; a barometer and a sealed reference channel on each logger hub are candidates too, the error model carries the barometric efficiency of each well, the drift of each logger, cable cross-talk and model error within stated ranges, and the network is chosen against the worst plausible error and reported with its certified optimality gap; a conductivity logger in at least one production well of each wellfield and a yearly campaign of laboratory samples paired 1, 3 and 7 days apart measure the time scale that decides crossings between samples; procurement and installation (T4.2), with one GNSS station, two corner reflectors, a seismic station and a water-table logger at the TAP landfall, under TAP's security rules; operation and maintenance (T4.3), in which every reading is corrected for air pressure and logger drift before use and the corrected residuals are checked for remaining correlation; seismic velocity as a water-table gauge (T4.4). Risk: procurement delays and limited site access; mitigation: early orders, inductions with security officers. Risk: lagged or uncertain barometric responses; mitigation: response functions estimated from the first three months by regression deconvolution, with their error carried into the robust design. Milestones: MS1 (Month 6), MS2 (Month 12).

WP5, Live demonstration (Months 22–36; NATO-country team). Steps: dry run (T5.1); twelve months of bulletins (T5.2); safe operating window and pumping rule (T5.3): the effect of each wellfield's pumping

on salinity is estimated with the hidden drought demand removed, by reading the recharge from the heads after adding back the drawdown measured in the pumping tests, and a calibrated check of every substitute is run before any effect is reported; the salinity under a drier recharge regime is computed from the calibrated state with conformal bands; the certified window, its closing year and safe pumping tests that sharpen it follow, and the pumping and drought rule is agreed with the utility and AMBU; post-event protocol (T5.4). Risk: no strong event occurs; mitigation: verification by back-testing and storylines. Risk: drawdown coefficients from the pumping tests are uncertain; mitigation: their uncertainty is carried into the effect as a stated bias band. Milestone: MS6 (Month 34).

WP6, Training, transfer and dissemination (Months 1–36; partner-country team): early-career researchers and a training school (T6.1), operator training (T6.2), open release (T6.3), transfer to a second coast (T6.4). Risk: the operators rely on the bulletins alone, or trained operator staff leave; mitigation: the system stays advisory beside the operators' own procedures, the training explains what a certified miss rate means, and at least two people are trained per operator. Milestone: MS7 (Month 36), the operators own and run the system: an operating procedure adopted for each asset, every bulletin of Months 34–36 issued by the operators without the project team, and a maintenance agreement signed by Month 33 (D6.2).

WP7, Management, data and risk (Months 1–36; coordinator): management and reporting (T7.1), data plan and access agreements (T7.2), risk register (T7.3). Risk: the operators delay access to well, port or landfall data; mitigation: access agreements by Month 3, with WP1 and the design of WP4 started on public and satellite data meanwhile. Risk: a release discloses sensitive information on critical infrastructure; mitigation: data classified in the data plan, asset-level results shared only with the operators and authorities, and publications reviewed by the operators before submission. Milestones: access agreements and data management plan (D7.1, Month 3); progress reports at the SPS milestones (D7.2, Months 6, 18 and 30); final report (D7.3, Month 36); the risk register reviewed at every gate.

After the project ends: Sustainability and role of end-user(s)

Implementation. The operators issue every bulletin themselves in Months 34–36, with at least two trained staff each, and sign a maintenance agreement (Month 33) that covers the running cost of about €10,900 per year for data, telemetry and spare sensors. Until it is in place, the Albanian university hosts the system. Operating procedures for each asset and a policy brief for the national authorities (D6.2) anchor the warnings in daily practice.

Next steps: extend the warnings to the other wellfields and coastal assets of Albania; offer the open software to coasts of Allies and partners (tested on a second coast in the partner country during the project); seek national and European funding to keep the network and the verification ledger running.

Role of the end-users. The water utility, the Water Resources Management Agency (AMBU), the port authority, the operator of the Trans Adriatic Pipeline (TAP) and the National Civil Protection Agency guide the project from the start: they give access to sites and records, agree the thresholds and tolerable miss rates (T3.1), take part in the dry run, co-sign the operational milestones and run the bulletins in live operation. The operators own the system after the hand-over; AMBU uses its results without procurement, ownership or maintenance obligations. End-users receive no SPS funds.

Risks and mitigation strategies

Risk description	Likelihood	Impact	Mitigating action
Operators delay access to well, port or landfall data; calibration and design start late	Medium	High	Letters of support at proposal stage; access agreements by Month 3 (T7.2, D7.1); data needs listed per operator
Historical head and chloride records are incomplete or of uneven quality	High	Medium	Data rescue from paper archives; quality flags; record errors carried as uncertainty in the assimilation
The coupled model fails its validation test on held-out data	Medium	High	Build in stages (flow, then salt, then compaction and liquefaction); test each against data before coupling

Risk description	Likelihood	Impact	Mitigating action
Certified alarms are too cautious: the false-alarm rate is unacceptable to the operators	Medium	High	Acceptable false-alarm rate agreed with each operator at T3.1; tolerances allocated by consequence; network designed to shrink margins
Conditions move beyond the observed record (drought, compound storm, post-seismic change) and weaken the guarantees	Medium	High	Extreme-aware margin term; storylines beyond the record in back-testing (T3.4); nonstationary extremes with land motion
No strong earthquake or surge occurs during the demonstration, so the guarantees are not tested on such events	High	Medium	Back-testing over the historical record, including the 2019 earthquake, and on storylines beyond it (T3.4); exceedance events per alarm counted first, since a 10% miss rate needs at least 29 events to verify (Methodology 5.7)
A missed alarm coincides with damage; trust and responsibilities are questioned	Low	High	Every bulletin states its certified miss rate; decisions stay with the operators; roles written into the bulletin protocol (T5.1)
Sensitive information on critical-infrastructure vulnerability is disclosed through open data or publications	Medium	High	Data classification in the data plan (T7.2); asset-level results shared only with operators and authorities; open release of methods, code and non-sensitive aggregated data; operators review publications before submission

All 22 risks, with owners and triggers, are in the Risk Assessment document.

Communication Plan

Activities: a public kick-off, a public start of live operation and a final event; a project website with public summaries in Albanian and English; a quarterly verification ledger showing, for each alarm type, events, misses, false alarms and the bound on the miss rate; at least six open-access papers and eight conference presentations; a policy brief for the national authorities; stakeholder workshops at kick-off, mid-term, end and for transfer.

Presenting the project to the public: the start of live warnings, the training school (Month 14) and the hand-over to the operators are natural moments for press releases and local media.

Likely photo and video subjects for the SPS Office (after security clearance by the operators): installing piezometers and GNSS stations, corner reflectors on the coast, the training school, a civil-protection exercise with the bulletins.

Budget link: the heading "Communication and Publications" (€18,000) covers open-access fees (€12,000), the website, translation and printing (€5,000) and public events (€1,000). Asset-level information is never published; every output passes a release gate.

Key Performance Indicators (KPIs)

500 words.

The project is on track when the following are achieved on time:

- D4.1 network design approved by the steering committee and the operators, with a certified optimality gap of at most 25% for the chosen network (Month 6, MS1).
- D4.2 network installed and streaming with at least 90% data availability, and the corrected heads within 3 cm root-mean-square of manual dips (at most half the raw error); D1.1 coastal motion atlas with a vertical velocity uncertainty of about 1–2 mm/yr at asset points (Month 12, MS2).
- D2.1 calibrated coupled model: held-out heads, chloride, InSAR and seismic velocity inside the 90% predictive intervals at close to the nominal rate, and for each wellfield a certified band at most 0.25 wide for the chance that chloride crosses its limit within the season (Month 18, MS3).
- D2.2 certified surrogate: error bound met on test runs; alarm prototype passes a dry run (Month 22, MS4, go/no-go for live operation).
- D3.2 back-testing: each alarm's miss rate at or below its target with a 95% confidence bound, crossings between samples included, under a protocol published with a DOI before the results (Month 27, MS5).

• D5.1 twelve months of live bulletins with every outcome logged, the miss rates consistent with their certificates, and the certified safe operating window reported for each asset, with every safe pumping test below its chloride test limit (Month 34, MS6).

• D6.1 open software and data with a DOI, reproduced by someone outside the team (Month 33); D6.2 operating procedures adopted and policy brief delivered (Month 36, MS7).

• Capacity: at least 20 trained operator and civil-protection staff, at least six early-career researchers, at least six open-access papers and eight conference presentations.

• Management: progress reports at each SPS milestone, spending within 10% of the plan per budget category, and the risk register reviewed at every gate.

311 words (limit 500)

Criteria for success

At most seven, SMART, weights summing to 100%.

Criterion	Relative weight (%)
Each alarm's miss rate in back-testing over the record since 2014 is at or below its certified target (indicatively 5–10%), with a 95% confidence bound (D3.2, Month 27)	25
Twelve months of live bulletins (Months 23–34) with every outcome logged and no verification test rejecting a certified miss rate (D5.1)	20
Held-out heads, chloride, InSAR and seismic velocity fall inside the model's 90% predictive intervals at a rate of at least 85% (D2.1, Month 18)	15
The operators issue all bulletins in Months 34–36 with at least two trained staff each, and the maintenance agreement is signed (MS7)	15
The monitoring network reaches at least 90% data availability over twelve months of operation (D4.2, T4.3)	10
The certified safe operating window and its closing year are reported for each asset and confirmed by the full model at the closing point, and every pumping test stays below its chloride test limit (T5.3)	10
The open release is installed and reproduces the synthetic test coast for a group outside the consortium (D6.1, Month 33)	5

List of abbreviations

AI: artificial intelligence; CF: Climate and Forecast conventions; CPT: cone penetration test; D: deliverable; DOI: digital object identifier; dv/v: relative change of seismic velocity; EDT: emerging and disruptive technologies; FAIR: findable, accessible, interoperable, reusable; GDPR: General Data Protection Regulation; GNSS: global navigation satellite system; InSAR: interferometric synthetic aperture radar; KPI: key performance indicator; MS: milestone; MYP: Multi-Year Project; NPD: NATO-country Project Director; PPD: partner-country Project Director; SPS: Science for Peace and Security; T: task; TRL: technology readiness level; WP: work package.

Optional upload: a file that helps to illustrate the project

Upload CERTAIN-COAST_Illustrative_Figures.pdf (ready): six pages, a table of contents and twelve figures (the coast as one system, the critical assets and NATO resilience, the method, the innovations in the warning chain, the gaps closed, the certified safe operating window, single-hazard horizons, the alarm margin, warnings between samples, the self-correcting network, milestones and gates, impact pathway).

8. Project Schedule & Budget

Project duration in months

36

Project schedule (upload)

PDF with a Gantt chart: duration and interdependence of each step, critical milestones, deliverables.

Upload CERTAIN-COAST_Project_Schedule.pdf (ready): Gantt chart with the duration of every task, the dependencies, the critical path, the milestones and the deliverables by SPS milestone. Project milestones: MS1 network design (Month 6), MS2 network streaming (Month 12), MS3 calibrated model

(Month 18), MS4 go/no-go for live operation (Month 22), MS5 miss rates confirmed (Month 27), MS6 twelve months live (Month 34), MS7 hand-over (Month 36).

Milestones overview

SPS reporting milestones; must match the "Info" tab of the budget workbook (at least one milestone every 12 months).

Event	Description	Months after kick-off
Kick-off	Project kick-off meeting	0
Milestone 1 (required)	First technical and financial report	6
Milestone 2 (required)	Second technical and financial report	18
Milestone 3 (required)	Third technical and financial report	30
Milestone 4 (optional)	Fourth technical and financial report	leave blank
Final report (required)	Final technical and financial report	36

Budget summary by item: NATO SPS funding requested (in EUR)

Indicative split by reporting period, assuming even spending within each project year (period ending at Milestone 1 = months 0–6, M2 = 6–18, M3 = 18–30, final = 30–36). Enter the final figures from your Excel workbook.

Budget category	M1	M2	M3	M4 (opt.)	Final	Total
Equipment	75,900	91,400	15,500	—	0	182,800
Training	2,100	14,800	18,400	—	5,700	41,000
Communication and Publications	1,125	3,750	7,875	—	5,250	18,000
Travel	6,900	14,700	17,100	—	9,300	48,000
Consumables	3,805	7,878	7,545	—	3,472	22,700
Other	5,000	8,000	6,000	—	3,000	22,000
Stipends	7,200	16,800	16,800	—	7,200	48,000
Co-financing of personnel	0	0	0	—	0	0
Total	**102,030**	**157,328**	**89,220**	**—**	**33,922**	**382,500**

Equipment

Major equipment, its role, approximate price, co-funding.

- Monitoring boreholes: drilling, casing, screens, sealing: €32,000. Piezometers where the network design (D4.1) ranks them highest; sealed to standard (R12)
- Pressure, conductivity and temperature loggers: €14,400. Heads and salinity in new and existing wells: the core data for intrusion and the water table
- Barometric loggers: €1,200. Barometric correction of the pressure loggers
- Telemetry units (mobile data loggers): €7,200. Near-real-time data for the live bulletins (WP5)
- GNSS stations: receiver, antenna, monument: €27,000. Vertical land motion at the wellfields and port of Durrës and the TAP landfall near Fier; ties InSAR to a global frame (T1.2)
- InSAR corner reflectors: €9,000. Stable radar targets on soft ground at the assets (R4)
- Seismic stations: three-component sensor and digitiser: €30,000. Continuous ambient noise for seismic velocity at the port of Durrës and the TAP landfall near Fier (T4.4)
- Seismic node rental, two campaigns: €8,000. Dense temporary coverage to calibrate velocity against the water table
- Fibre-optic sensing campaign: rental and processing (conditional): €15,000. Released only if fibre access is agreed by Month 9 (R9); otherwise moved to seismic nodes
- Power and enclosures: solar panels, batteries, locked boxes: €7,000. Autonomous and secure operation of the field stations (R11)
- Field chemistry instruments: multiparameter meter, sampling pump: €6,000. Conductivity and temperature in the field; sampling for chloride
- Spare loggers and sensors: €6,000. Replacement within two weeks after damage or theft (R11, R21)
- Computing server with GPU: €12,000. Thousands of coupled-model runs and training of the certified surrogate (T2.4)
- Redundant data server and backup storage: €5,000. Operational redundancy for the live bulletins (R17)

• Software licences: €3,000. Processing tools without an open-source equivalent; the project itself releases open software

Co-funding: [existing equipment and facilities of the institutions; operators provide access to wells, port and landfall].

Training

SPS does not fund fellowships; training periods are short.

• Young-researcher exchanges, two months each: €21,000. Partner-country early-career researchers work on site with the data and the operators (T6.1)

• Training school: venue, materials, trainee travel: €15,000. Hands-on school with the coast kit in Month 14 (T6.1)

• Operator and civil-protection training sessions: €5,000. Staff able to run the system unaided (T6.2)

Travel

• Consortium meetings (kick-off, mid-term, final): travel and subsistence: €21,600. Six people from the two countries at three meetings

• Stakeholder workshops: kick-off, mid-term, final, transfer: €12,000. Thresholds, dry run, hand-over and transfer to other coasts (Dissemination Plan)

• Conference presentations: €14,400. At least eight presentations of the results (Dissemination Plan)

Co-financing of personnel

Up to 25% of the personnel cost.

None requested: the institutions pay all salaries (see Non-NATO funding). The management hours of the NPD (€15,000) are under "Other", as the MYP Handbook (April 2026) requires. [If salary co-financing is wanted after all: per person and month at most the share of time on the project, capped at 25%, times the monthly gross salary; in total up to about one third of the grant; a CV for each recipient.]

Other major expenses (optional)

• Shipping and customs for equipment: €4,000. Delivery of sensors to the site

• Insurance of field equipment: €3,000. One year of cover per year of operation

• Man-hours for the management of the project (NATO-country Project Director only): €15,000. Coordination, reporting and financial records, with time records; SPS category "Other"; about 4% of the total

Communication and Publications (€18,000) and Consumables (€22,700) are detailed in the budget workbook and the Communication Plan above.

Budget summary by item: non-NATO funding contribution (in EUR)

Estimate what the institutions and nations contribute.

Budget category	What it covers	Amount from other funding sources
Equipment	Existing equipment used by the project: field instruments, laboratory instruments and servers of the institutions; the existing wells, piezometers and loggers of the water utility; existing seismic stations. Value: [share of the replacement value or depreciation over the 36 months].	[€ amount]
Training	Venues, rooms and trainers' time of the institutions for the training school and the operator training; operators' staff time as trainees is under "Other".	[€ amount]
Communication & Publication	[Institutional open-access agreements or press offices, if any; otherwise 0].	[€ amount]
Travel	[Institutional travel funds used for the project, if any; otherwise 0].	[€ amount]
Consumables	[Laboratory chloride and major-ion analyses done by the water utility or an institution laboratory at its own cost, if agreed; otherwise 0].	[€ amount]
Stipends	None expected (stipends of the early-career researchers are requested from SPS).	[€ amount]
Co-financing	Personnel cost paid by the institutions: all salaries of the Co-Directors and researchers for their time on the project (Section 3). SPS co-finances no salary in this version.	[€ amount]

Budget category	What it covers	Amount from other funding sources
Other	Overheads of the institutions (offices, administration, IT) at their own rates; staff time of the five end-users; data provided under agreements (Durrës tide-gauge record of ASIG, records of the national seismic stations, records of the water utility).	[€ amount]

Amounts, salaries and the personnel worksheet: CERTAIN-COAST_Non-NATO_Contribution.docx.

Non-NATO funding

The SPS grant pays only project-specific costs. The participating institutions and the end-users contribute the rest:

- Salaries of the Co-Directors and researchers, paid by their institutions for about 23.4 person-months over 36 months (NPD 10.8, PPD 7.2, researcher of the Institute of Geosciences 5.4): [€ amount]. No salary co-financing is requested from SPS.
- Overheads of the institutions (offices, laboratories, IT, administration) at their own rates: [€ amount].
- Existing equipment and facilities used by the project: [field and laboratory instruments, servers] of the institutions: [€ amount].
- End-users, who receive no SPS funds: access to the wells, the port and the landfall; existing wells, loggers and records; staff time for thresholds, the dry run, the live bulletins and training (about 9.0 person-months): [€ amount].
- Data at no cost to the project: Copernicus Sentinel-1 radar images (free and open), the Durrës tide-gauge record of ASIG and the records of the national seismic stations under data agreements [not valued / € amount].
- [Other national or institutional co-funding, only if confirmed: source and amount].

Total non-NATO contribution: [€ amount], [x]% of the SPS grant (€382,500).

How do you want to share the financial rules of the NPD's institution?

Choose "Link to a website": https://upt.edu.al/wp-content/uploads/2023/02/Statuti-PDF-English.pdf (Statute of the Polytechnic University of Tirana, English version published by UPT: Chapter IV "Properties, financing and services", Articles 24–25; Article 10, the Board of Administration approves the financial regulation). [If the UPT finance office provides its internal regulation on administration, finance and accounting, choose "File upload" and upload that PDF instead.]

9. End-Users

At least one end-user; all based in a NATO or eligible partner country; end-users receive no SPS funds.

First name	Family name	Title	Job title	Institution	Postal address	Country	Telephone	E-mail
[First name]	[Family name]	[Title]	[Job title]	[Durrës water utility]	[Postal address]	[Country]	[Telephone]	[E-mail]
[First name]	[Family name]	[Title]	[Job title]	[Water Resources Management Agency (AMBU)]	[Postal address]	[Country]	[Telephone]	[E-mail]
[First name]	[Family name]	[Title]	[Job title]	[Durrës Port Authority]	[Postal address]	[Country]	[Telephone]	[E-mail]
[First name]	[Family name]	[Title]	[Job title]	[Trans Adriatic Pipeline AG (TAP), Albania branch]	[Postal address]	[Country]	[Telephone]	[E-mail]
[First name]	[Family name]	[Title]	[Job title]	[National Civil Protection Agency]	[Postal address]	[Country]	[Telephone]	[E-mail]

End-User(s) commitments (upload)

One PDF with all signed letters, each on the end-user's letterhead (drafts in the Letters of Commitment document; the end-users sign their own letters). For governmental end-users, a letter of support from the ministry or agency.

10. Project attachments

Brief overview of participating institutions (upload)

One PDF, at most one page per institution: capabilities, resources, facilities, websites. Use CERTAIN-COAST_Institutions_Overview.docx: [fill the yellow fields, delete the pages of institutions that do not take part], then save it as PDF.

SPS MYP Agreement for Joint Research (upload)

NATO template, signed by all Co-Directors and a representative of each institution.

Do you want to include the Intellectual Property Rights (IPR) Agreement (or Waiver) now?

Yes (recommended): upload the signed agreement as one PDF. Draft: CERTAIN-COAST_IPR_Agreement_Draft.docx (NATO sample, Option B, open release of software and data, security release check), with a one-page waiver as alternative. The form and the Handbook allow it after approval, but the 2027-1 call lists it among the documents of a complete proposal.

Additional files (optional)

Upload CERTAIN-COAST_Technical_Annex.pdf (7 pages): how the methods work, the preparatory tests, the gates with their fallbacks and the six new contributions, consistent with this form. [When the CVs of the Co-Directors are filled in, upload instead one PDF that merges the annex and the CVs, with a table of contents at the start, as the form asks.]

Additional information (optional)

The technical annex under Additional files summarises the methods and their preparatory tests; the full methodology, state of the art, work packages, risk assessment and data management plan are available on request.